Harmonic Oscillations in Coupled Pendulums

PenPress Benchmark Series • Assignment 1

1. Theoretical Foundation

Consider two identical pendulums of mass m and length l suspended from a rigid horizontal beam. The pendulums are coupled by a linear spring of spring constant k attached at distance d below the pivot points. In the small-angle approximation, the equations of motion decouple into symmetric and antisymmetric normal modes.

Equation 1: $$\ddot{\theta}_1 + \omega_0^2 \theta_1 + \kappa (\theta_1 - \theta_2) = 0$$

Equation 2: $$\ddot{\theta}_2 + \omega_0^2 \theta_2 - \kappa (\theta_1 - \theta_2) = 0$$

2. Normal Mode Frequencies

By taking the sum and difference of the coordinates, we define the in-phase coordinate eta = theta_1 + theta_2 and the out-of-phase coordinate zeta = theta_1 - theta_2. The eigenfrequencies of the normal modes are given by:

Equation 3: $$\omega_1 = \sqrt{\frac{g}{l}}, \quad \omega_2 = \sqrt{\frac{g}{l} + \frac{2kd^2}{ml^2}}$$

3. Experimental Procedure

1. Displace both pendulums by identical angle theta_0 = 5 deg in the same direction and record period T_1.

2. Displace both pendulums by opposite angles theta_1 = -theta_2 = 5 deg and record period T_2.

3. Release pendulum 1 from theta_0 while pendulum 2 remains at rest, measuring the beat frequency omega_b.